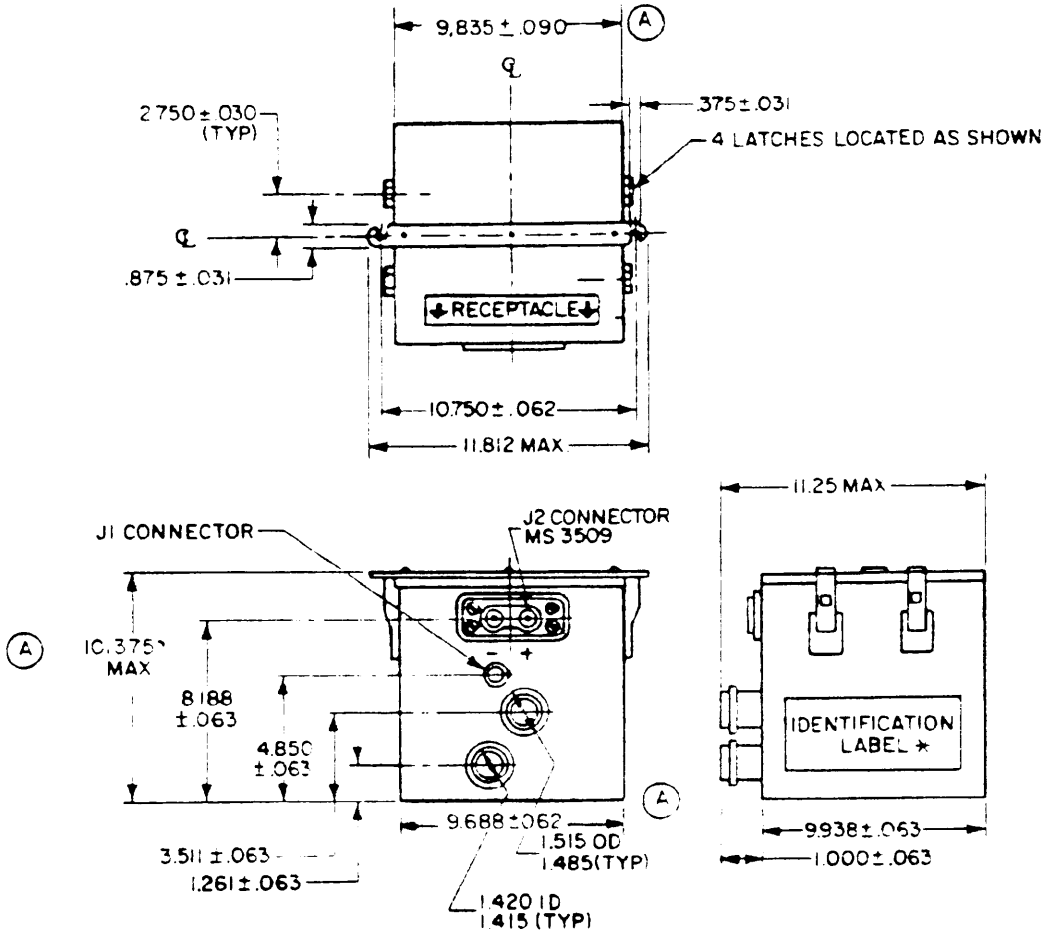
MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM FORCED-AIR COOLED, CHARGE CONTROLLED, 24-VOLT, 30 AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery assembly described herein shall consist of this document and the latest issue of MIL-B-81757.



* CAUTION LABEL REQUIRED IN SAME LOCATION, OPPOSITE SIDE OF CONTAINER.

FIGURE 1. Battery assembly.
P/N M81757/12-1 and P/N M81757/12-2

AMSC N/A
FSC 6140

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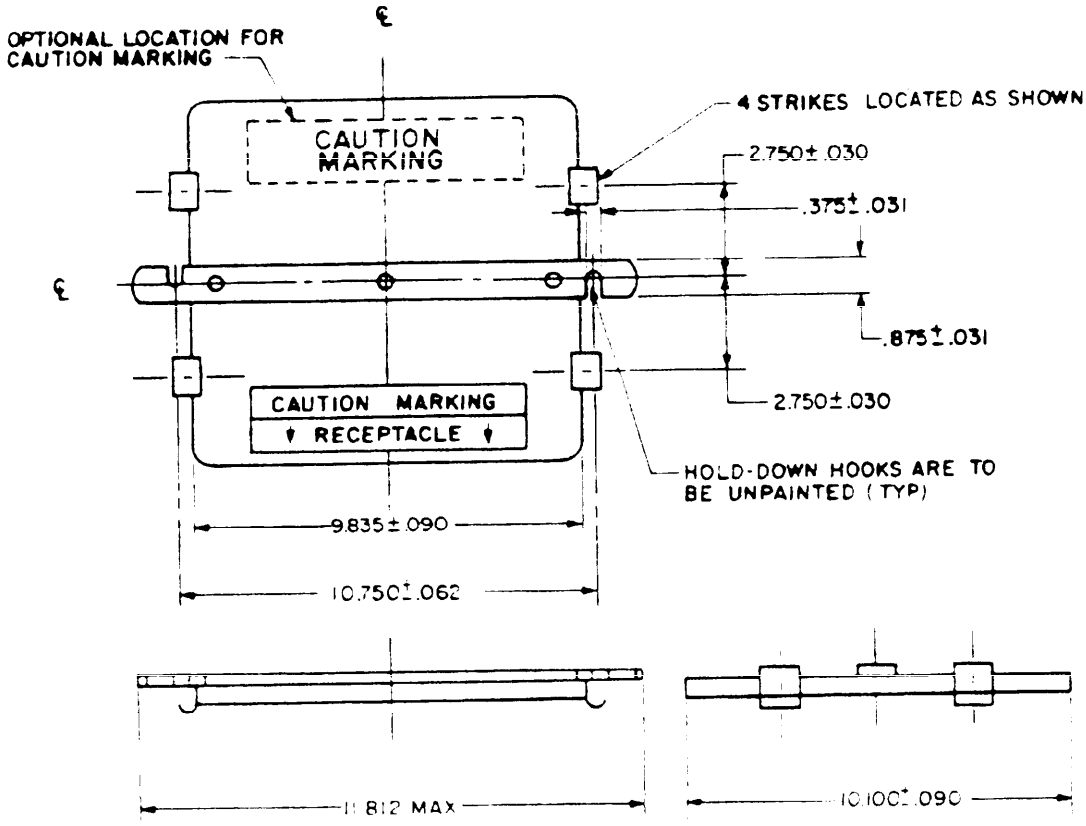


FIGURE 2. Cover.

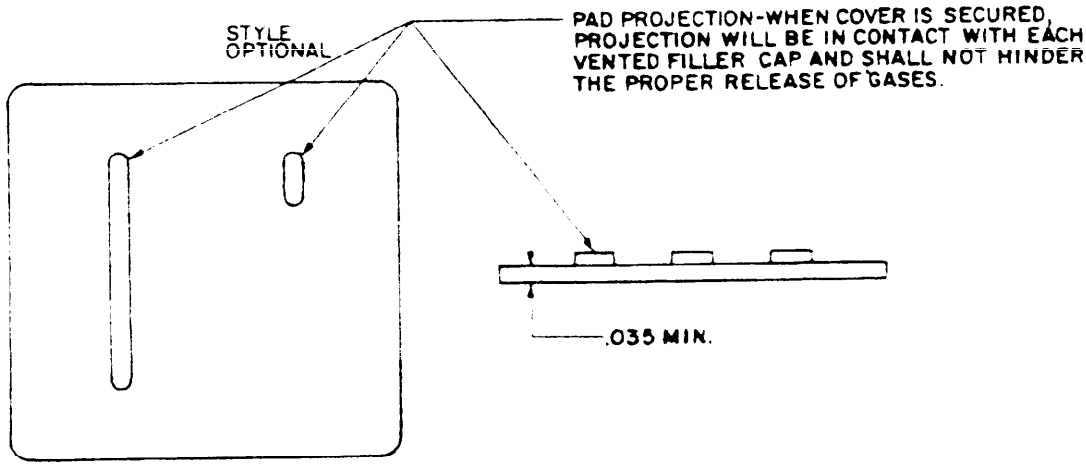


FIGURE 3. Cover hold-down pad.

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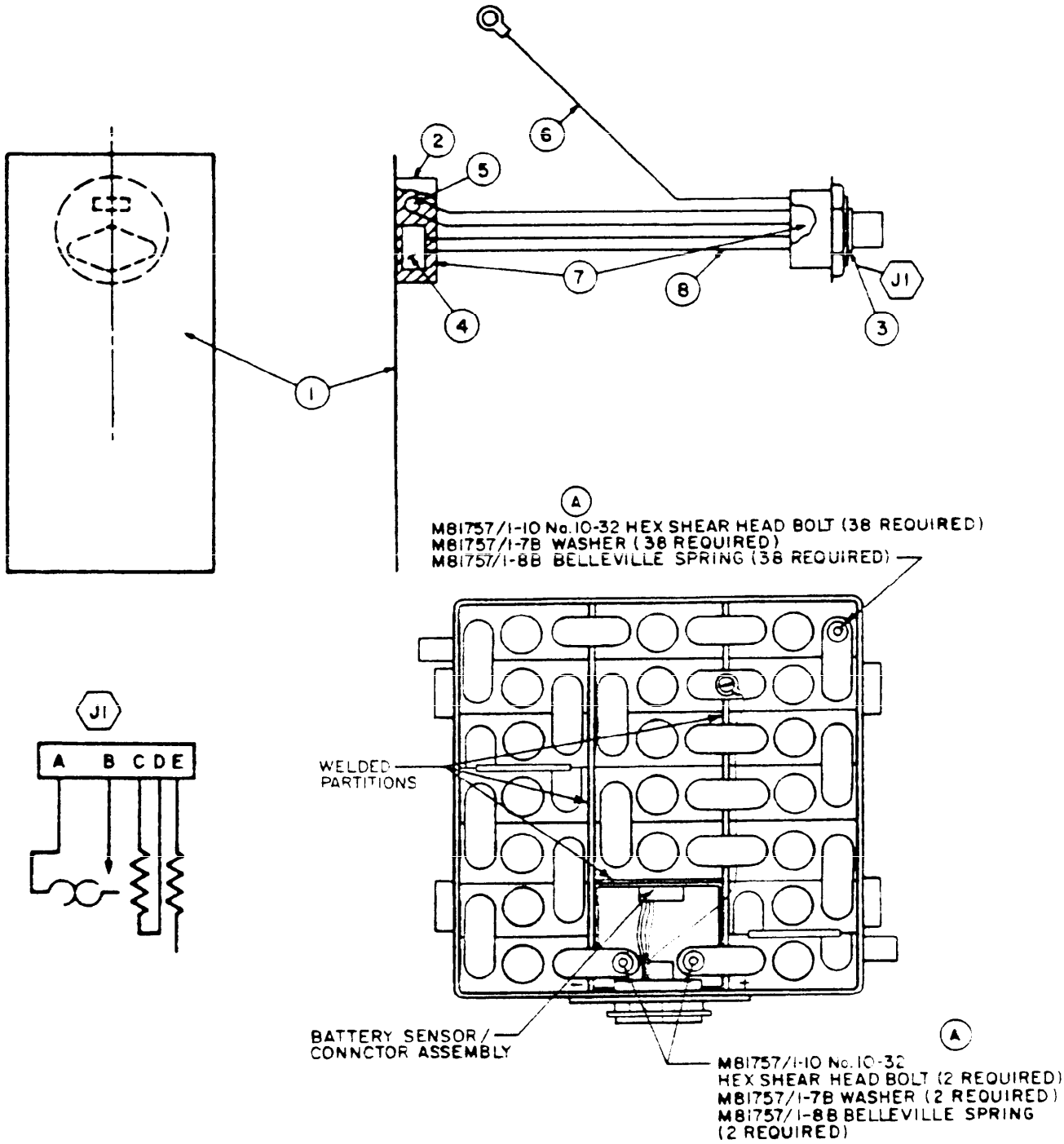


FIGURE 4. Cell arrangement and battery sensor/connector assembly.

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NOTES:

All M81757/12 batteries shall have the same cell arrangement.

1. Battery M81757/12-2 shall include a battery sensor/connector assembly capable of withstanding all environmental conditions to which the battery is subjected without damage or degradation of performance. The battery sensor plate shall be in thermal contact with the front surface of cell 16 (counting from negative terminal). No air flow shall be allowed to flow between the surfaces of cell 16 and the battery sensor plate. The battery sensor/connector assembly shall consist of:

   a. One temperature sensor plate ① with component mounting cavity ② , installed into the battery as shown.

   b. One auxiliary receptacle ③ , having ten contacts, type D38999/44WC98SN, conforming to MIL-C-38999, shall be mounted to the battery container as shown.

   c. One thermal switch ④ , in accordance with M24236/1-1565, having normally open contacts, which close at 71° ± 2.8°C and reopen at 62.8 ± 2.8°C. The thermal switch shall be certified to withstand a minimum of 50,000 operating cycles at 30 VDC and 2.0 amperes without the switch's opening and closing temperature values changing more than ± 2.8°C. The thermal switch shall be potted with the switch leads connected to pins A and B of the battery auxiliary connector.

   d. One 1/4-watt wire-wound resistor ⑤ , in accordance with MIL-R-39005, valued at 3,000 ohms ± 1 percent at 25°C, and shall have temperature coefficients of:

      T.C. = 3,900 ± 30 PPM/°C below 25°C
      T.C. = 4,100 ± 30 PPM/°C above 25°C

      The resistor shall be potted with the resistor leads connected to pins C and D of the battery auxiliary receptacle.

   e. One center tap voltage lead ⑥ , having a nickel-plated terminal lug, shall be connected to the positive terminal of cell 10 (counting from battery negative). The center tap voltage lead shall include a 2,000 ohm, 1/8-watt resistor, in accordance with MIL-R-39017, and be connected to pin E of the battery auxiliary connector.

   f. All electrical connections made to the battery auxiliary connector and battery sensor assembly shall be potted ⑦ with a compound that will not support flame, absorb moisture, or be damaged by battery electrolyte.

   g. All wire ⑧ used for electrical connections shall be #20 AWG, in accordance with MIL-W-16878, teflon insulated, and capable of withstanding the temperature and exposure to battery electrolyte without damage.

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REQUIREMENTS:

(A) 1. The receptacle will be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle terminal block shall be screw size 10-32. Receptacle shall conform to MS3509.

2. All dimensions are prior to epoxy coating except: Overall length - 11.812 maximum, overall width - 11.250 maximum, and overall height - 10.312 maximum.

(A) 3. Method of mounting latches and strikes shall be optional. They shall conform to Corbin Cabinet Lock Division, American Hardware, Item Number 15834, or Nielson Latch number C-833314-2-SS-302 or equivalent.

(A) 4. Spacers, if used, shall be polyethylene sheet in accordance with ASTM STD D4066-82, Composition A, Type I and Type III, minimum 0.015 inch thick, color neutral.

5. Latches and strikes shall not be epoxy-coated.

6. The battery voltage shall be 24 volts (nominal).

7. The battery weight shall be 88 pounds maximum when filled.

8. The battery shall contain 19 cells.

9. The rated capacity at 1C rate shall be 30 Ah/1 Hr/80°F/18.0V.

10. Length and width of cover pad and spacer shall conform to the inside dimensions of the applicable cover.

11. The length, width and thickness of cover pad projections shall provide sufficient pressure on the vented filler caps to hold down the battery cells during vibration.

12. The quantity and location of cover projection pads shall be determined by the vented filler caps.

13. All cells within a battery shall be from the same production lot.

(A) 14. Cell and receptacle removal shall be prevented by non-removable fasteners, M81757/1-10 hex, shear-head bolt.

15. The cell vented filler cap shall be in accordance with MIL-B-81757/2

16. Battery part number M81757/12-1 shall conform to MIL-B-81757 except as modified by the following:

3.5.7 Cells. Delete.

3.5.7.1 Mountings. Delete.

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3.6.5.2.1 Cell terminal marking. Delete last three sentences.

3.6.5.2.2 Cell identification. Delete.

3.6.5.2.3 Battery identification. Each battery shall have its identification marking silk screened with an epoxy overlay, or equal, that is impervious to 1.3 ± 0.025 specific gravity KOH electrolyte, in accordance with applicable specification sheet. Marking shall be in white print in accordance with MIL-M-13231. The marking shall show the following:

Information Required	Example
Battery, Storage	Battery, Storage
Military Battery Part No.	M81757/12-1
Replaces (include replacement data from applicable battery specification sheet)	Replaces D8565/2-1
National Stock Number	NSN 6140-01-***-****
Voltage (Nominal)	24 Volts
Maximum Weight	Maximum Weight 88 Lbs
Military "C" Rated Capacity	(30 AH/1 Hr/80°F/18V)
Contract Number	***
Date Code (see 3.6.5.2.5)	Date Code **
Manufacturer's Part Number	***
(A) Manufacturer's Serial Number	***
Warranty Expiration Date	041787

3.6.5.2.4 Battery caution and receptacle markings. These markings, in accordance with applicable specification sheet, shall be in white printing in accordance with MIL-M-13231. They shall be impervious to 1.3 ± 0.025 specific gravity KOH electrolyte. They shall be located as shown on the applicable specification sheet and read as follows:

↓RECEPTACLE↓	0.50 ± 0.03 Inch High Letters
"CAUTION"	0.50 ± 0.03 Inch High Letters
THE CELLS IN THIS BATTERY ARE NOT TO BE REMOVED.	0.375 ± 0.03 Inch High Letters
THIS BATTERY SHALL BE SERVICED ONLY IN AN AUTHORIZED NICKEL-CADMIUM BATTERY SHOP.	0.375 ± 0.03 Inch High Letters

17. 3.6.17 Cooling effectiveness test. The batteries, when tested in accordance with 4.6.24, shall meet the following:

a. Exhibit a decline in battery temperature of 38.9°C in a maximum of 100 minutes.

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b. The pressure difference between the inlet air and the outlet air shall not be greater than 50.8 mm water equivalent.

18. 4.6.24 Cooling effectiveness test (qualification). Batteries submitted for test shall be subjected to the cooling effectiveness test as follows:

a. Battery temperature monitoring shall be accomplished by submerging a protected thermocouple at least 5 millimeters below the surface of the electrolyte in the cell vent opening of cell number 15 (counting from negative terminal).

b. The battery shall be charged in accordance with 4.5.1.4.2 of MIL-B-81757. Following a 2-hour stand, the electrolyte of each cell shall be adjusted.

c. The battery, with cover in place, shall be placed in a temperature chamber at 60 ± 1.1°C until the temperature of the electrolyte in cell number 15 (counting from negative terminal) reaches the ambient temperature of the chamber.

d. Following the stabilization of battery temperature, the battery shall be removed from the temperature chamber and cooling air shall be applied to the inlet of the battery. The air shall be applied at 7080 cu cms/sec at a temperature of 21.2 ± 1.7°C.

e. The temperature of the battery shall be monitored by the thermocouple in cell number 15 (counting from negative terminal), and the time shall be measured until the battery temperature decreases to 38.9°C.

f. The battery shall meet the requirements of 3.6.17.

NOTES:

1. MIL-B-81757/12(AS) supersedes DOD-B-8565/2.

2. All dimensions are in inches.

International standardization. Certain provisions of this specification sheet are the subject of International Standardization Agreement (ASCC AIR STD 12/15 and STANAG 3514). When revision or cancellation of this specification sheet is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including department standardization offices, if required.

Preparing Activity:
Navy - AS
(Project No. 6140-N644)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-81757/12A(AS)	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM FORCED AIR COOLED, CHARGE CONTROLLED, 24-VOLT, 30 AMPERE-HOUR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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